

What is information need? What are the different types of information need? (VCFC)

Def: The requirement to interested information of users.

1. **Visceral need (本能需求):** the actual, but unexpressed need for information : “a vague sort of dissatisfaction ... probably inexpressible in linguistic terms”.
2. **Conscious need (有意识的需求):** the conscious within-brain description of the need: “an ambiguous and rambling statement”. i.e. “I know what I’m looking for but I don’t know how to say it”.
3. **Formalised need (正式需求):** the formal statement of the question
4. **Compromised need (折衷需求):** the question as presented to the information system

分辨文档是否有phrase

注意事项:

- phrase中关键词出现次数和间隔
- phrase中关键词位置（是否在第一个等）

Example(23Q1.b)

Which document(s) could contain the phrase “the same day at the same time”

```
1 | <same: 41825;  
2 | 1: 120, 124, 167;  
3 | 2: 9, 10, 13;  
4 | 3: 121, 162;  
5 | 4: 4, 101, 105, 106;  
6 | 5: 1, 5, 88, 888, 889;  
7 | ...>
```

- 答案: 1
- 原因:
先看same在phrase中出现位置: 相隔三个单词, 145都符合, 但4中106也是same, phrase中没有相邻same, 4排除。5中第一个same在句首, 不符合phrase中same位置, 5排除。

解释Boolean searches, simple ranking and reranking based on machine learning为什么对IR系统重要

- **Boolean search:** Very quick initial select, narrow massive data by using boolean operators (AND/OR/NOT).
- **Simple ranking (BM25):** Also fast, perform Light weight & Simple but Effective Ranking,

- **Reranking based on machine learning:** Slow, reorders the top- k BM25 results using machine learning.

BM25三个principles

- Term Frequency (**TF**): A way of ranking how important term is in a document collection.
- Inverse Document Frequency (**IDF**): Count of the number of times the term appears in the document.
- Document length: BM25 normalises the length of document in calculations.

$$sim_{BM25}(d_j, q) \sim \sum_{k_i \in d_j \wedge k_i \in q} \frac{f_{i,j} \times (1+k)}{f_{i,j} + k \left((1-b) + \frac{b \times len(d_j)}{avg_doclen} \right)} \times \log \left(\frac{N-n_i + 0.5}{n_i + 0.5} \right)$$

TF
IDF
DL

Stemming and Lemmatisation 定义, Adv & Disadv

Technique	Definition	Advantages	Disadvantages	Example
Stemming	Process of reducing words to their root form	Fast processing	May overstem (cut too much)	"computing", "computer" → "comput"
Lemmatization	NLP technique that converts words to their base/dictionary form (lemma)	More effective	Slower processing	"running" → "run"

Cranfield Paradigm

The **Cranfield Paradigm** is a foundational experimental methodology in information retrieval that provides a standardized framework for evaluating retrieval systems. It involves:

1. A standard corpus used for all queries
2. A set of standard queries
3. A set of relevance judgements

Its advantages include repeatability, objectivity, and efficiency.

非英语中tokenising会遇到的问题

French: L'ensemble → one token or two?

German: German noun compounds are not segmented

Chinese and Japanese have no spaces between words

Arabic is basically written right to left, but with certain items like numbers written left to right.

为什么representation对IR很重要？

Information retrieval is defined as dealing with the “representation, storage, organization of, and access to information items”

- **Storage and Organisation:** There are important considerations in terms of storing documents also. They must be organised so as to be **quickly accessible**. This is particularly important with large document collections.
- **Access:** The key attribute of an IR system is to allow users to access the information in question. This should happen in a timely and efficient manner.
- **Representation:** Represent textual documents in mathematical form so retrieval is faster.

Adversarial Information Retrieval是什么，举两个例子（MHCSED）

Def: Violating search engine guidelines and rules to try and get an artificial advantage:

Technique	Definition	Purpose
Meta Tags Abuse	Manipulating <code><meta></code> tags (e.g., keyword stuffing) for early search engines.	Artificially boost visibility for targeted searches.
Hidden Text	Concealing keywords via CSS/font color matching background.	Inflate term frequency (TF) without user visibility.
Cloaking	Serving different content to crawlers (e.g., via User-Agent detection).	Trick crawlers into indexing keyword-rich content while users see normal pages.
JavaScript Deception	Using <code><noscript></code> or JS redirects to show hidden text to crawlers.	Get crawlers to index hidden keywords while redirecting users.
PageRank Exploitation	Creating link farms or comment spam for artificial backlinks.	Manipulate PageRank scores to improve rankings.
Doorway Pages	Creating multiple optimized pages for different queries, all funneling to one target.	Maximize keyword coverage while avoiding on-page keyword dilution.